

Improved Greedy Algorithm for Protein Structure Reconstruction

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Abstract: This article concerns the development of an improved greedy algorithm for protein structure reconstruction. Our stochastic greedy algorithm, which attempts to locate the ground state of an approximate energy function, exploits the fact that protein structures consist of overlapping structural building blocks that are not independent. Application of this approach to a series of 16 proteins with 50–250 amino acids leads to predicted models deviating from the experimental structures by 0.5 Å RMSD using an RMSD-based energy function and within 1.5 to 4.8 Å RMSD using a Go-based energy function. The Go-based results are significant because they illustrate the strength of combining structural fragments and stochastic greedy algorithms in capturing the native structures of proteins stabilized by long-range interactions separated by more than 30 amino acids. These results clearly open the door to less computationally demanding solutions to predict structures from sequences.

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Introduction

Ab initio protein structure prediction from the optimization of a scoring function and without using any templates from the Protein Data Bank (PDB) is an important issue in biological science.¹ Although the thermodynamic hypothesis is well documented, that is, the native or experimental structure of most proteins is associated with the ground state of the free energy, finding this global minimum on the free energy surface is a challenging computational problem. First, the energy model must be designed such that the global minimum for each amino acid sequence is reached for its X-ray diffraction or NMR solution structure. Second, the protein folding problem is NP-complete because the number of conformations grows exponentially with the number of variables.²

Much work has been done over the last years to overcome this sampling problem. Replica exchanges or parallel tempering simulations where exchanges between simulation copies at different temperatures are permitted,³ and ensemble dynamics, which consists of submitting in parallel several tens of independent series at constant temperature, each series consisting of hundred coupled Langevin molecular dynamics (MD) simulations,⁴ have opened new perspectives in folding polypeptides, but such studies require very large computer facilities. Other heuristic methods have also

been developed to find good, but not necessarily guaranteed, optimal solutions. These include simulated annealing,⁵ tabu search,⁶ genetic algorithm,⁷ combined tabu search with genetic algorithm,⁸ chain growth strategies,⁹ Monte Carlo (MC) simulations using biased probability functions of the main chain (ϕ , ψ) torsional angles,¹⁰ kinetic-MC methods aimed at locating the transition states between the conformations (e.g., the activation-relaxation technique¹¹), and multicanonical MC simulations.¹² From tests, these heuristic methods have provided reasonable solutions for polypeptides but remain to be tested on proteins (larger than 50 amino acids).

The effectiveness of all methods depends upon their ability to avoid local energy conformations and exploit the basic structure of the problem. Building on these notions, recent methods have exploited the fact that protein structures involve the succession of recurrent conformations of few residue length. As a result, the problem to be solved is twofold: first, predict the preferred conformations of fragments (these can be secondary structures), and then assemble them into a three-dimensional model using various techniques.^{7,9,13} The most promising of such approaches is RO-

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SETTA, which identifies fragments of known structures of three and nine residue length with local sequences similar to the target sequence, and assembles them using simulated annealing or genetic algorithms.^{13,14}

Recent work has moved one step further and has taken advantage of the fact that the conformation of the overlapping short fragments along the sequence is not independent. Using a Hidden Markov Model (HMM), Camproux et al. learnt a structural alphabet (HMM-SA) consisting of 27 distinct prototype conformations (letters or Structural Building Blocks, SBB) as well as the transitions between the letters, and reconstructed the structure of proteins with an accuracy of 1 Å α -carbon root mean square deviation (RMSD).¹⁵ To do so, they used the optimal series of SBBs describing each protein structure, an overlapping scheme to assemble the prototypes and the greedy algorithm described by Park and Levitt,¹⁶ and Kolodny et al.¹⁷ In their work, the reconstruction was guided by minimizing the RMSD from the experimental structure.

Greedy algorithms, which grow a polypeptide chain one fragment after another, have a long history in structure prediction or docking problems.^{9,16–20} Clearly, greedy algorithms using an RMSD-based energy function are rather straightforward and need to be revisited for less stringent energy functions that do not restrict the search at each incremental position. In this work, we present an improved greedy algorithm and demonstrate its efficiency in generating folded protein structures using a nonoptimal trajectory of SBBs under the minimalist Go-potential. The Go-potential,²¹ although not optimal, provides some realistic energy model of protein folding and allows us to address the problem of long-range interactions stabilizing the native structures. As a first step, we discuss the impact of several variations of the algorithm on a series of 16 proteins with 50–250 amino acids using nonoptimal trajectories of SBBs.

Materials and Methods

Encoding Protein Structures into a Sequence of SBBs

For each protein, we define a specific set of fragments by taking advantage of the possibility of typing its conformations as a series of SBBs, so as to mimic what could be expected as a result of prediction from the sequence. HMM-SA describes the conformations of the overlapping fragments of four residue length of a protein structure. Thus, for a protein of length $L + 3$ the resulting encoded structure is a trajectory of SBBs of length L .

Definition of the Fragments Associated with Each SBB

Having encoded 1429 protein structures from PDB, we have extracted and identified for each SBB a set of representative fragments using a standard dynamic clustering procedure based on a minimal dissimilarity threshold (here a maximal RMSD from the centroid of the class). Hence, the number of prototype fragments associated with each SBB depends on its conformational variability. Using a threshold of 0.5 Å, it varies from 1 to 23, and results in a collection of 155 fragments for the 27 states, for an average of 4.6 prototypes per letter.

Identification of the SBBs Describing the Local Conformation at Each Position of the Structure

It is possible to encode each protein structure as a series of letters of the alphabet using the Viterbi or the forward-backward algorithms, for example. The Viterbi algorithm identifies the best state sequence (or optimal trajectory) among all the possible paths in $\{1, \dots, 27\}^L$ by dynamic programming, taking into account the Markovian dependence between the consecutive states. The forward-backward algorithm computes, for each position, the probability that the structure is represented by each of the 27 states.

Here, we used the forward-backward approach to escape from the optimal description of the structure—that is, one letter only to describe each position—and to introduce fuzziness in the local conformations. At each position of the structure, we have retained the letters of the alphabet associated with a probability higher than a threshold of 0.01. This number of letters retained at each position depends on the local structure of the proteins and varies from 1 to 6 (among a possible total of 27 letters) for our protein set. The corresponding profile of SBBs compatible with the structure will be referred to as a fuzzy trajectory. During reconstruction, we thus consider at each position i all the fragments (see the section Definition of the Fragments Associated with Each SBB) associated with each of the letters retained, that is, n_i fragments.

Fragment Assembly

Going from the N-terminus toward the C-terminus, the fragments are assembled by superimposing the first three alpha-carbons of the new fragment onto the last three of the previously built structure of length l . The superimposition is performed by a best fit technique using the quaternions.²² This results in a new structure of length $l + 1$, where the fourth alpha-carbon of the new fragment is inserted. In order to define a reversible procedure allowing the exchange of one fragment by another, or to reverse the direction of the assembly (see sections below), we memorize the best fit coordinates of all the alpha-carbons of the new fragment instead of only those of its fourth alpha-carbon, and we use these coordinates to generate the best fit coordinates of the next superimposed fragment at position $i + 1$. Hence, exchanging the fragment k at position i by a fragment k' and re-exchanging fragment k' by fragment k will lead to the same coordinates.

Search Complexity

Considering a protein of length $L + 3$, that is, of L fragments of four residue length, and given that each position i among L is associated with n_i possible fragments (see previous section), the number of possible reconstructions for the whole protein is $\prod_{i=1}^L n_i$. Following the convention set by Park and Levitt,¹⁶ the complexity of the search (the average number of states per residue) is: $\sqrt[L+3]{\prod_{i=1}^L n_i}$. For instance, for one protein of seven residue length, that is, described by four letters or SBBs (e.g., *AXPJ*), if the number of representative fragments associated with the four letters is 1, 2, 5, and 4, respectively, the complexity is: $\sqrt[7]{1 * 2 * 5 * 4}$, that is, 1.69. Let us consider the fuzzy trajectory and that position 1 is now described by letters $[A, B, a]$, position 2 by letters $[V, X]$, position 3 by letters $[P, Q, R]$, and position 4 by letter $[J]$. If

the number of fragments associated with the letters $A, B, a, V, X, P, Q, R, J$ is 1, 1, 1, 3, 2, 5, 3, 6, and 4, respectively, the number of possible fragments for the four positions is now 3, 5, 14, and 4, respectively, and the complexity is $\sqrt[7]{3 * 5 * 14 * 4}$, that is, 2.61. We emphasize that although the fuzzy trajectories restrict the size of the conformational space, they do not drive the search to the correct solution. We show below that the basic greedy algorithm generates non-native and even random topologies for most targets using the Go energy model and the fuzzy trajectories.

Greedy Algorithm

Greedy algorithms perform the incremental construction of the structure starting from the N-terminus and ending at the C-terminus. Clearly, the number of possible reconstructions for a protein of length $L + 3$ is far too large to be explored exhaustively. The idea behind basic greedy algorithms is that, at each of the i positions, and after generating all the possible extensions of the structure, only a maximum of H conformations (heap size) among all the conformations generated are retained for the next step. This process is repeated until the C-terminal extremity is attained. The algorithm efficiency depends on the parameter H . The results presented here are obtained by setting H to 3000, a value similar to that employed in a previous study,¹⁷ and allowing a good trade-off between efficiency and rapidity. Such a basic procedure does not warrant, however, that the global optimum is reached. In this study, we consider several variants of this basic procedure.

Stochastic Greedy Algorithm

Efficiency of such a search algorithm is firstly related to its ability to escape from local minima. To address this issue, we introduce some randomness in the selection of the H best conformations. Given a current number of C_i conformations at position i , we retain only the H_b energetically best conformations, and randomly select the remaining $H - H_b$ conformations among the $C_i - H_b$ conformations of lowest energy. For example, if we consider all the letters of the alphabet at position i and use a heap size of 3000, this results in a number of 465,000 (3000×155) conformations. Let H_b be 1000, and this implies that we select the remaining 2000 conformations among the 464,000 other conformations.

Iterated Greedy Algorithm

One problem associated with an incremental construction of protein structure is that the selection of the conformations does not take into account the entire sequence, except when the extremity is reached. Clearly, we cannot ignore that long-range interactions may drive the algorithm in a different basin of the energy landscape. In order to address this issue, we introduce an iterated procedure on the whole greedy search. All iterations except the first one are performed on structural candidates describing the complete protein. In this process, going from N-terminus toward C-terminus, and having assembled a new fragment, we systematically reposition the C-terminus part of the structural candidate. This is possible because the assembly of the fragments is reversible (see the section Fragment Assembly). Here, we do not attempt to optimize the number of iterations and use two iterations.

Forward and Backward Growth

This consists of building the protein either from the N-terminus toward the C-terminus (forward growth) or from the C-terminus toward the N-terminus (backward growth). We recognize that the degree of sampling might be more complex for ab initio optimizations without any knowledge of the native contacts, for instance by starting the construction from arbitrary points, but this is beyond the scope of the present article.

Prefiltering

This consists of performing independent greedy runs on medium sized parts of the structure, and learning the fragments for exclusive use during the complete greedy search. Here, we have considered a moving window of 20 amino acid length, with a step of 15 amino acids, and for each step, we perform one forward and one backward search. We use a heap size of 1000 and a stochastic selection of 300 candidates. We emphasize that such a procedure changes the complexities only slightly, but it may eliminate some prototypes having incompatible geometries with good structures.

Simplified Energy Function

All calculations use the minimalist $C\alpha$ representation and the simplified Go energy function expressed by:

$$E_{\text{Go}} = \sum_{j>i} \Delta_{ij} B_{ij} + E_{\text{rep}}, \quad (1)$$

where B_{ij} is -1 kcal/mol and Δ is unity when residues i and j form a negative contact, and zero otherwise. A contact is defined native in the all-atom experimental structure (all hydrogens included) when any atoms of two nonconsecutive side chains ($j - i \geq 4$) are less than $3 \text{ \AA}$ from each other. A native contact is defined in the generated structures when its $C_\alpha - C_\alpha$ distance is $< \lambda$ times its value in the experimental structure. In this work, λ is set to 1.3,²³ and the results are not strongly dependent on the exact cutoff distance (nearly identical results for $\lambda = 1.2$, data not shown). The excluded-volume potential, E_{rep} , is set to 2 kcal/mol if two $C\alpha$ atoms separated by at least six amino acids are less than $3.5 \text{ \AA}$.

Results

The improved greedy algorithm was tested on a series of 16 proteins adopting various topologies. Table 1 describes their PDB codes, numbers of amino acids, and CATH structural classifications²⁴ and reports the complexity of the problems to be solved using the optimal and fuzzy SBB trajectories. We see that fuzziness in the local conformations increases complexity by a factor of 1.5 on average at each position and the total number of possible conformations by 1.5^{L+3} for a protein of length $L + 3$. All calculations were performed on Linux Xeon PC, and a structural reconstruction for a 100-residue protein took on the order of 20 min using the most complex version of the algorithm.

Table 1. Protein Targets and Complexities of the Various Trajectories.

PDB ID	Len	CATH	C-opt	C-fuzzy
1csp	67	β	4.37	6.28
1ctf	68	α/β	2.48	4.45
1hlb	157	α	2.95	5.90
1pdo	129	α/β	2.95	4.72
1pgb	56	α/β	2.88	4.54
1shg	57	β	4.51	6.30
1tfe	142	α/β	2.41	3.98
1timA	247	α/β	3.65	6.75
1ubq	76	α/β	3.44	4.78
2acy	98	α/β	3.44	6.28
2bby	69	α	2.65	4.92
2ci2	65	α/β	3.54	4.08
2ife	91	α/β	5.17	5.82
2lzm	164	α	2.59	4.57
3chy	128	α/β	2.73	4.40
6pti	56	Few	4.02	6.12

For each PDB code, we give the number of amino acids (Len), the CATH description, and the complexity associated with the optimal (C-opt) and fuzzy (C-fuzzy) trajectories as described in the text. 6pti is classified as having few secondary structures by CATH.

Basic Greedy Algorithm

The results of the basic greedy algorithm using the RMSD-based evaluation function are first evaluated. For the 16 constructed proteins generated in a single run, we find an average RMSD of 0.93 ± 0.2 Å using the optimal trajectories and 0.55 ± 0.15 Å using the fuzzy trajectories. This shows that the global reconstruction of protein structures from knowledge of the $C\alpha$ positions—this corresponds to minimization of the RMSD-based function—is rather independent of the number of degrees of freedom, although higher complexities slightly improve the reconstructed proteins. This has already been discussed in detail for a large set of proteins

using our library of fragments and other libraries^{17,25} with the RMSD-based function.¹⁵ In contrast, higher complexities become problematic using less stringent functions (e.g., Go energy model), because the search is not restricted at each incremental position as in the reconstruction under the RMSD-based criterion. We show, however, that our modified version of the greedy algorithm can solve this issue.

Improved Greedy Algorithm: Short Chains

In what follows, we examine the impact of the variations in the greedy algorithm on the generated structures of 10 short chains (<100 amino acids) using the Go-energy model and the fuzzy trajectories. For each protein, we list in Table 2 the energy of the experimental structure (E_{ref}), the lowest-energy predicted structure with its RMSD from the native structure using the basic greedy algorithm, and the iterated greedy algorithm with both the forward and backward chain processes. Overall, we see that the second iteration of the greedy algorithm leaves invariant most targets, but makes significant contributions on four models: by moving the 2ci2 and 1ctf models to lower energy regions (variation in energy from -56 to -63 using GF1 and GF2 for 2ci2), or by moving the 1ubq and 2bby models to lower energy and RMSD regions. Specifically, using the second iteration of the greedy algorithm and forward growth of the chain, we see that one target (1ubq) deviates by 2.0 Å RMSD from the experimental structure with an energy very similar to the optimal value, but all remaining targets deviate within 3.2 Å (1pgb)–10.8 Å (2acy) RMSD from the solutions. On the other hand, growing the protein chain from the C-terminus to the N-terminus (backward growth) fails for 50% of the targets, but improves the solutions for the remaining targets and notably for 2bby (1.96 Å RMSD and optimal energy). These results show that the specificity of the experimental topology has to be taken into account: the N-terminal end can be disordered and the C-terminal stabilized by secondary structures (e.g., β -hairpins in 2bby and 2ife), and vice versa.

Table 2. RMSD (in Å) and Energy (kcal/mol) of the Predicted Structures for the 10 Short Chains Using the Basic and Iterated Greedy Algorithms, and the Forward and Backward Chain Models under the Go-Energy Function.

ID	E_{ref}	GF1	E	GF2	E	GB1	E	GB2	E
1csp	-78	7.70	-63	7.70	-63	8.07	-62	8.26	-65
1ctf	-79	9.69	-46	6.56	-61	10.67	-51	10.00	-53
1pgb	-62	3.24	-55	3.24	-55	2.84	-55	2.30	-59
1shg	-73	5.39	-61	5.39	-61	3.56	-62	3.56	-62
1ubq	-95	4.97	-82	2.04	-93	7.35	-68	4.40	-85
2acy	-130	11.14	-79	10.83	-82	14.82	-68	14.92	-73
2bby	-65	4.98	-64	4.68	-65	4.32	-64	1.96	-65
2ci2	-70	4.00	-56	4.08	-63	5.80	-54	3.79	-60
2ife	-112	5.90	-96	5.82	-97	4.01	-108	4.01	-108
6pti	-53	6.12	-44	6.12	-44	9.60	-31	9.60	-31

GF1 and GF2 are the first and second iterations of the basic greedy algorithms using the forward (F) growth model, whereas GB1 and GB2 refer to the iterations using the backward (B) growth model. In contrast to the first iteration, the second iteration considers the whole protein. The targets that are significantly improved are in bold.

Table 3. RMSD (in Å) and Energy (in kcal/mol) of the Predicted Structures for the 10 Short Chains Using Two Iterations of the Algorithm, Stochastic Effects, and the Forward and Backward Chain Models under the Go-Energy Function.

ID	E_{ref}	GF	E	GB	E	SF-R2	E	SF-R1	E	SB-R2	E
1csp	-78	7.70	-63	8.26	-65	7.71	-63*	7.70	-63*	4.20	-75
1ctf	-79	6.56	-61	10.00	-53	2.74	-79	4.94	-64	1.16	-78
1pgb	-62	3.24	-55	2.30	-59	2.80	-62	2.39	-62	2.18	-62
1shg	-73	5.39	-61	3.56	-62	2.97	-65	3.47	-64	2.65	-71
1ubq	-95	2.04	-93	4.40	-85	2.58	-95	1.94	-95	2.47	-95
2acy	-130	10.83	-82	14.92	-73	3.41	-117	8.86	-93	9.91	-89
2bby	-65	4.68	-65	1.96	-65	3.87	-65*	3.43	-65*	2.50	-65*
2ci2	-70	4.08	-63	3.79	-60	1.51	-70	1.66	-70	2.61	-70
2ife	-112	5.82	-97	4.01	-108	3.04	-110	3.41	-111	4.40	-106
6pti	-53	6.12	-44	9.60	-31	2.13	-53	1.78	-52	1.93	-53

GF and GB are the greedy (G) algorithms using the forward (F) and backward (B) growth models, whereas SF and SB are their stochastic (S) variants using 1000 (-R1) or 2000 (-R2) randomly chosen conformations in the heap at each position. For simplicity, the results of SB-R2, being very similar to those of SB-R1, are not given. The targets that are not sensitive to stochastic effects are marked by asterisks.

We now report in Table 3 the results using two iterations of the algorithm, stochastic effects, and other variations of the algorithm from a total of five runs. Addition of randomness in selecting the best structural candidates at each position within the heap is an important factor in improving the solutions. With stochastic effects included, forward growth results in improved quality for eight targets and provides predicted structures within 4.0 Å RMSD for all targets but 1csp (7.7 Å RMSD). On the other hand, stochastic effects and backward growth improve the quality of nine targets and allow 70% of the targets to deviate by less than 2.7 Å RMSD with optimal or near-optimal energies. Figure 1 shows the plot of energies as a function of the RMSD for 10 runs on the 1shg and 2acy targets. Clearly, the stochastic procedure, by keeping a reservoir of high-energy conformations in the heap, allows a more realistic assembly if the chain momentarily lacks native contacts, that is, when the native topology is characterized by two consecutive native contacts separated by 10–20 amino acids. In this work, we do not attempt to optimize the number of random conformations. We see, however, that this parameter is topology-dependent. Changing the number of random conformations from 1000 to 2000 improves the quality of four targets during forward growth and six targets during backward growth.

Finally, we examine the additional effect of prefiltering the fragments prior to the complete stochastic search. For simplicity, we only report in Table 4 the targets that are improved upon prefiltering. We see that prefiltering improves the 1ubq and 2ife targets using forward growth, and 50% of the targets using backward growth.

Table 4 also summarizes the best generated models for our 10 short chains and provides strong evidence of the interest of using all of the variants of our greedy algorithm to generate the lowest energy solutions. This is not too surprising because all targets vary in secondary structure composition and topology. Overall, we see that 60% of the targets, namely 1ctf, 1pgb, 1ubq, 2bby, 2ci2, and 6pti deviate within 2.2 Å from their experimental structures, although they have optimal or near-optimal energies. This result, consistent with MD simulations at room temperature—RMS fluc-

tuations of the backbone are on the order of ~ 2.0 Å—indicates that the lower bound for optimal reconstruction is 2.0–2.5 Å using our Go-energy model based on the positions of the $C\alpha$ atoms. The other four targets deviate from 2.7 (2ife) to 3.4 Å (2acy) with predicted energies differing by 1% (2ife, 1shg), 4% (1csp), and 10% (2acy) from their optimal values. These deviations in energy, expected from a heuristic procedure, reveal, however, that the predicted topologies are native. Figure 2 shows the superposition

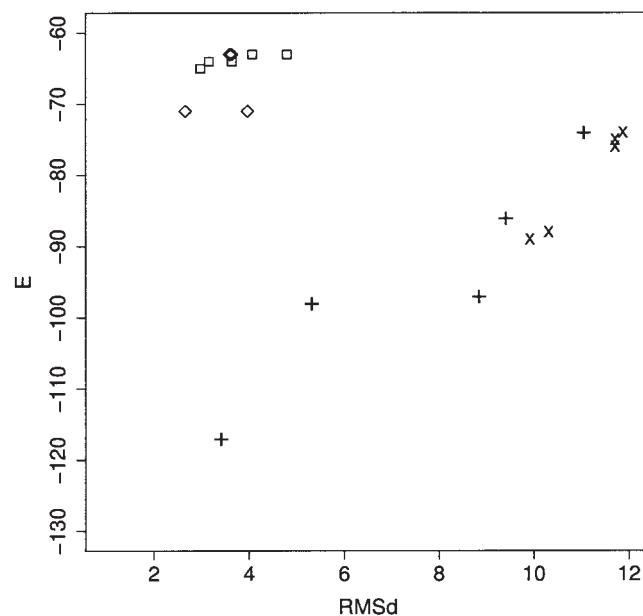


Figure 1. Pseudo energy (in kcal/mol) as a function of the RMSD (in Å) for five runs using the SF and SB stochastic variants. 2acy: + denotes the SF runs, × the SB runs. 1shg: squares denote the SF runs, diamonds the SB runs.

Table 4. Effects of Prefiltering on RMSD (in Å) and Energy (in kcal/mol) of the 10 Short Predicted Protein Structures. Comparison between the best predicted and experimental models.

ID	E_{ref}	SF-R2-p	E	SB-R2-p	E	Best	E
1csp	-78			3.14	-75	3.14	-75
1ctf	-79			2.02	-79	1.16	-78
1pgb	-62			2.19	-62	2.18	-62
1shg	-73			3.21	-72	3.21	-72
1ubq	-95	1.86	-95			1.86	-95
2acy	-130					3.41	-117
2bby	-65			2.04	-65	2.04	-65
2ci2	-70					1.51	-70
2ife	-112	2.68	-111			2.68	-111
6pti	-53					1.93	-53

SF-R2 are SB-R2 are the stochastic (S) variants, using the forward (F) and backward (B) growth models, 2000 randomly chosen conformations (-R2) in the heap at each position, and prefiltering the fragments (-p). For clarify, only the targets that are improved upon prefiltering are given. The last two columns describe the best predicted model obtained using all variants of our algorithm.

of the experimental and predicted structures for the 1csp (RMSD of 3.1 Å), 1shg (RMSD of 3.2 Å), and 2acy (RMSD of 3.4 Å) targets. The greatest structural divergence between the predicted and experimental 1csp or 1shg structures is found at positions where the experiments show the lowest structural determination (e.g., the first and last two residues). Or it is found at positions lacking Go-type contacts with the remaining part of the protein; for instance in 1csp, the residues 35–38 and 42–43 located within the loop (33–45) are free to move under our Go-energy model. On the other hand, analysis of the contact map of the 2acy predicted model shows an incomplete list of interactions between the β -strands 1 and 4, and between β -strands 2 and 3 (see Fig. 2c). We anticipate, however, that such an energy and structural difference might be partially solved by sampling the conformational space using kinetic-MC methods and more complex energy functions.¹¹

Improved Greedy Algorithm: Long Chains

Table 5 reports the best generated models generated by both our basic and improved greedy algorithms for six long chains (120–250 amino acids). The basic greedy algorithm with one iteration and the forward chain growth model performs reasonably for the 1pdo and 3chy targets although their energies are not optimal, but

generates random topologies for 1tfe, 2lzm, 1timA, and 1hlb with RMSDs of 6.4, 10.2, 12.5, and 15.1 Å, respectively. In contrast, the improved greedy algorithm generates models for 1pdo, 2lzm, and 3chy with optimal (experimental) energies and RMSDs within 2.4–2.9 Å. The best model for the 1hlb deviates by 4.7 Å RMSD from the experimental structure and has a predicted energy differing by 1% from its optimal value. Finally, the best model for 1timA, consisting of 247 amino acids, deviates by 4.8 Å RMSD from experiment, but has the experimental topology, and remarkably, closes the barrel with native contacts between the two extremities (Fig. 3).

Conclusions

We have shown that our improved greedy algorithm allows us to reproduce 16 protein structures within 0.5 Å RMSD using the RMSD-based energy function and within 1.5 to 4.8 Å RMSD using the Go-based evaluation function. These Go-based results are significant because they clearly illustrate the strength of combining HMM-derived structural fragments and stochastic greedy algorithms in capturing the native structures of proteins stabilized

Table 5. Comparison between the Best Predicted and Experimental Models for the Six Long Chains Using the Basic (GF1) and Improved Greedy (SB-R2-p and SF-R2-p) Algorithms.

Experimental ID	E_{ref}	Basic greedy RMSD	E	Improved greedy RMSD	E
1hlb	-154	15.1	-90	4.70	-152
1pdo	-166	3.9	-160	2.45	-166
1tfe	-183	6.4	-159	2.62	-183
1timA	-263	12.5	-190	4.84	-243
2lzm	-204	10.2	-148	2.56	-204
3chy	-171	4.1	-157	2.97	-171

RMSD in Å and energy in kcal/mol.

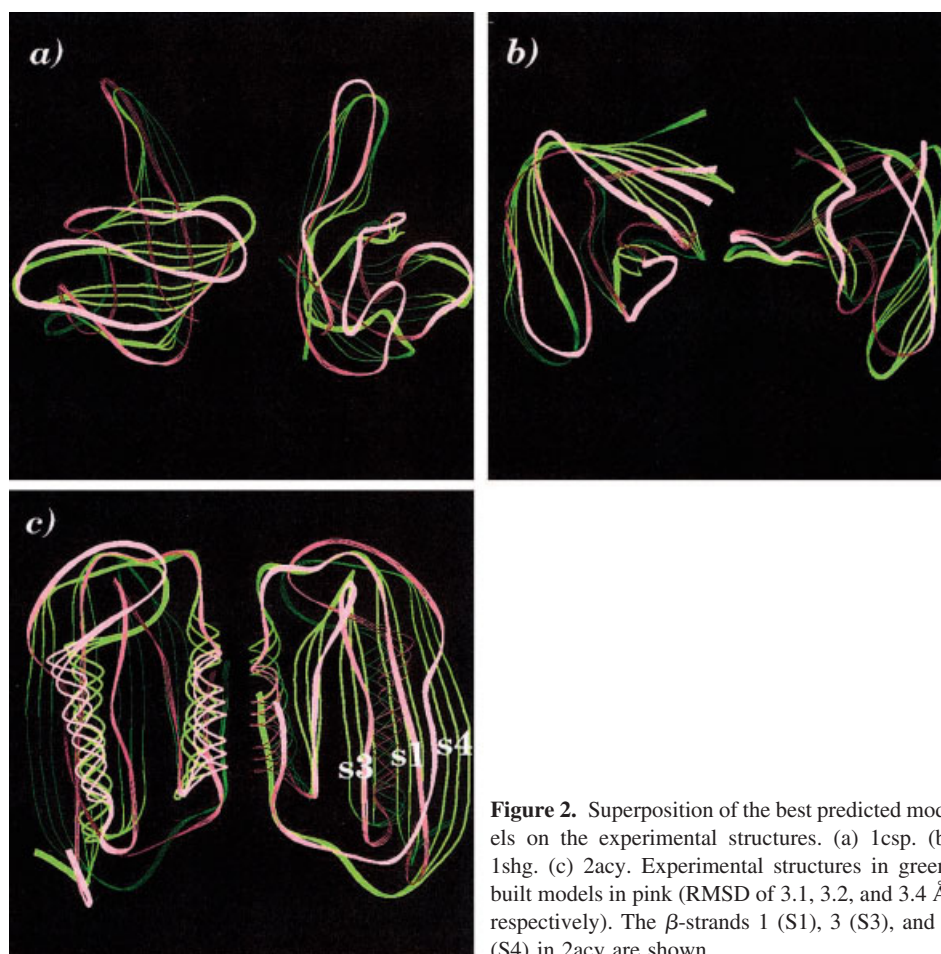


Figure 2. Superposition of the best predicted models on the experimental structures. (a) 1csp. (b) 1shg. (c) 2acy. Experimental structures in green, built models in pink (RMSD of 3.1, 3.2, and 3.4 Å, respectively). The β -strands 1 (S1), 3 (S3), and 4 (S4) in 2acy are shown.



Figure 3. Superposition of the best predicted and experimental structures for 1timA (247 amino acids). (Top) Experimental structure in green and predicted model in pink with all residues shown. (Below) Experimental structure in yellow and predicted model in blue with all residues shown and the helix spanning 17–36 deleted. The two terminal strands S1 and S8 are indicated.

by long-range interactions. In this context, the tests on the 2acy and 1timA targets are very difficult. In 2acy, the β -strand 1 (L6-K16) is stabilized within the native structure by interactions with the β -strands 3 (V47-P54) and 4 (H74-V85), that is, by interactions separated by 40 and 70 amino acids, respectively. In 1timA, the two extremities are predicted in native contact, although they are separated by more than 200 amino acids. Interestingly, our improved greedy algorithm also performs very well using fuzzier trajectories that encode the secondary structure information (as determined by the STRIDE program²⁶) into its corresponding SBB and use the full size alphabet of 27 SBBs for all the remaining positions.²⁷ This suggests that our methodology clearly opens the door to less computationally demanding solutions to predict structures from sequences, even in the ab initio context.

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